INCH-POUND

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SUPERSEDING
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21 May 1992

COMMERCIAL ITEM DESCRIPTION

MANDRELS, MACHINE, EXPANDING

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers expandable mandrels which are used for supporting workplaces between centers for machining operations.

2. CLASSIFICATION. Unless otherwise specified in the acquisition order (see 7.1), the mandrels shall be classified by the following sizes:

2.1 Size. See table I.

3. SALIENT CHARACTERISTICS

3.1 Design. The mandrels shall be a cylindrically shaped tool consisting of a solid arbor having a specific taper over the major portion of its length and one or more expanding sleeves with an inside taper which fits the tapered portion of the arbor. The arbor shall be centered on each end with a milled flat on one end for a holding device. When the sleeve is moved along the length of the arbor, it shall contract or expand as determined by the location on the arbor. The plane of the outside surface of the sleeve shall be parallel with the horizontal center line of the arbor. The mandrels shall be similar to figure 1.

3.2 Tolerance and finish. The arbor centers shall be machined to a standard angle of 60 degrees ± 30 minutes. The surface roughness average (Ra) of the arbors shall be not greater than 20 microinches using a 0.030 inch cutoff when measured in accordance with ASME B46.1. The maximum allowable runout shall be not greater than 0.001 inch total indicator reading at the grind diameter. The sleeves shall be finished internally to fit the arbor and shall provide uniform expansion to any size in the range of diameters listed in table I.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia (DSCP), ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&inspecomments@dla.mil . Since contact information can change, you may want to verify the currency of this address information using ASSIST Online database at https://assist.daps.dla.mil .

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Table I. Sizes and dimensions.

Size	Sleeve Expansion range 1/ (inches)	Arbor length 2/ (inches)	Sleeve length 2/ (inches)
1	3/8 to ½	4-1/2	1-1/4
2	1/2 to 9/16	5	2-1/2
3	1/2 to 11/16	5-3/4	1-1/2
4	9/16 to 21/32	6	2-3/4
5	21/32 to 3/4	6	2-3/4
6	11/16 to 1	9-1/4	2-1/2
7	3/4 to 7/8	7	3-1/4
8	13/16 to 1	9-1/4	2-1/2
9	7/8 to 1	7	3-1/4
10	1 to 1-3/8	9	4
11	1 to 1-9/16	12-1/4	3-1/4
12	1-1/4 to 1-5/8	9	4
13	1-1/2 to 2	11-1/2	5
14	2 to 2-3/4	14-1/2	5-1/2
15	2-3/4 to 3-3/4	17	6
16	3-3/4 to 5	20	7
17	5 to 6-1/2	24	8

1/ Where a range is shown, the dimensions shall be not greater than the lowest figure to not less than the highest figure.
2/ Tolerance of arbor length shall be ±1/2 inch and sleeve length ±1/4 inch.
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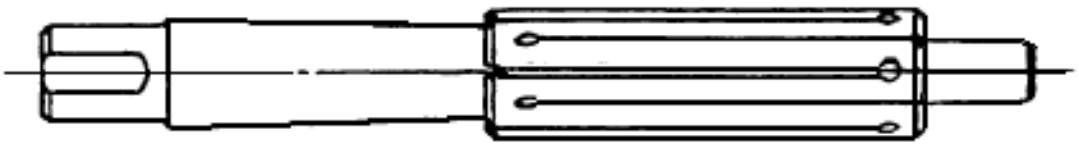

FIGURE 1. Mandrel, expanding type.

3.3 Hardness. The arbors shall have a hardness of 51 to 62 and the sleeves shall have a hardness of 38 to 47 on the Rockwell “C” scale when measured in accordance with ASTM E18.

3.4 Identification marking. Each mandrel furnished shall be permanently marked with the sleeve size or manufacturer’s part number and the manufacturer’s name or trademark of such known character that the source of the manufacturer may be identified.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING.

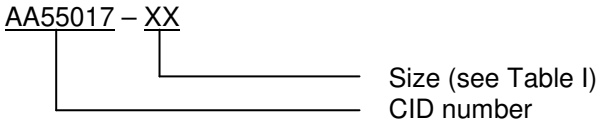
6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.1).

7. NOTES.

7.1 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. Size and quantity required.
c. Packaging requirements (see 6.1)

7.2 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



7.3 Sources of documents.

7.3.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001 or online at http://www.arnet.gov.far .

7.3.2 ASME standards. Copies of ASME standards may be obtained from the ASME International, Three Park Avenue, M/S 10E, New York, NY 10016-5990.

7.3.3 ASTM standards. Copies of ASTM standards may be obtained from ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959 or online at www.astm.org .

7.4 Key words.

Arbor
Cylindrical
Taper

7.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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